JESS Thermodynamic Database v8.9

Chemical species in reactions with Gold

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
2SHSuccin-3 Thiomalate ion				
-3	---	106	C(4)H(3)O(4)S(1)	147.125
Acetic-1 Acetate ion				
-1	71-50-1	558	C(2)H(3)O(2)	59.0446
Ag(s) Silver; Silver, cubic				
0	7440-22-4	182	Ag(1)	107.870
Ala-1 Alanine ion; L-Alanine ion; L-2-Aminopropanoate ion				
-1	---	802	C(3)H(6)N(1)O(2)	88.0861
As(s) Arsenic; Arsenic, rhombohedral; Arsenic, grey/gray; Arsenic, alpha				
0	7440-38-2	179	As(1)	74.9220
Asn-1 Asparaginate ion; L-2-Aminobutanedioate 4-amide ion; L-beta-Asparaginate ion; Aspartate beta-amide ion; Altheinate ion; Asparamide ion; Agedoite ion				
-1	---	581	C(4)H(7)N(2)O(3)	131.111
Asp-2 Aspartate ion; Aminosuccinate ion; Asparagate ion; 2-Aminobutanedioate ion; L-2-Aminobutanedioate ion				
-2	63-71-8	776	C(4)H(5)N(1)O(4)	131.088
Au-1				
-1	---	1	Au(1)	196.970

doi: 10.5281/zenodo.7700024

Au+1
Gold(I) ion

1	20681-14-5	110	Au(1)	196.970

Au+1_2SHSuccin-3

-2	---	1	C(4)H(3)Au(1)O(4)S(1)	344.095

Au+1_Acetic-1

0	---	2	C(2)H(3)Au(1)O(2)	256.015

Au+1_Acetic-1(2)

-1	---	2	C(4)H(6)Au(1)O(4)	315.059

Au+1_Br-1

0	---	1	Au(1)Br(1)	276.874

Au+1_Br-1_(s)
Gold(I) bromide; Gold monobromide

0	---	3	Au(1)Br(1)	276.874

Au+1_Br-1_OH-1

-1	---	3	Au(1)Br(1)H(1)O(1)	293.881

Au+1_Br-1(2)

-1	---	5	Au(1)Br(2)	356.778

Au+1_Cl-1

0	---	2	Au(1)Cl(1)	232.423

Au+1_Cl-1_(g)

0	---	1	Au(1)Cl(1)	232.423

Au+1_Cl-1_(s)
Gold(I) chloride; Gold monochloride

0	---	2	Au(1)Cl(1)	232.423

Au+1_Cl-1_OH-1

-1	---	4	Au(1)Cl(1)H(1)O(1)	249.430

Au+1_Cl-1(2)

-1	---	8	Au(1)Cl(2)	267.876

Au+1_Cl-1(3)

-2	---	1	Au(1)Cl(3)	303.329

Au+1_Cl-1(4)

-3	---	2	Au(1)Cl(4)	338.782

Au+1_CN-1(2)

-1	---	5	C(2)Au(1)N(2)	249.005

Au+1_Cys-2

-1	---	3	C(3)H(5)Au(1)N(1)O(2)S(1)	316.108

Au+1_Cys-2(2)

-3	---	1	C(6)H(10)Au(1)N(2)O(4)S(2)	435.246

Au+1_H+1_Cys-2(2)

-2	---	1	C(6)H(11)Au(1)N(2)O(4)S(2)	436.254

Au+1_I-1

0	---	1	Au(1)I(1)	323.870

Au+1_I-1_(s)

Gold(I) iodide; Gold monoiodide

0	---	5	Au(1)I(1)	323.870

Au+1_I-1(2)

-1	---	8	Au(1)I(2)	450.770

Au+1_NH3_Br-1_(s)

0	---	1	Au(1)Br(1)H(3)N(1)	293.905

Au+1_NH3_Cl-1_(s)

0	---	1	Au(1)Cl(1)H(3)N(1)	249.454

Au+1_NH3_I-1_(s)

0	---	1	Au(1)H(3)I(1)N(1)	340.901

Au+1_NH3_S2O3-2

-1	---	3	Au(1)H(3)N(1)O(3)S(2)	326.119

Au+1_NH3(2)

1	---	3	Au(1)H(6)N(2)	231.031

Au+1_NH3(2)_Br-1_(s)

0	---	1	Au(1)Br(1)H(6)N(2)	310.935

Au+1_NH3(2)_Cl-1_(s)

0	---	1	Au(1)Cl(1)H(6)N(2)	266.484

Au+1_NH3(2)_I-1_(s)

0	---	1	Au(1)H(6)I(1)N(2)	357.931

Au+1_NH3(3)_Br-1_(s)

0	---	1	Au(1)Br(1)H(9)N(3)	327.966

Au+1_NH3(3)_I-1_(s)

0	---	1	Au(1)H(9)I(1)N(3)	374.962

Au+1_NH3(4)_Br-1_(s)

0	---	1	Au(1)Br(1)H(12)N(4)	344.996

Au+1_NH3(6)_Br-1_(s)

0	---	1	Au(1)Br(1)H(18)N(6)	379.057

Au+1_NH3(6)_I-1_(s)

0	---	1	Au(1)H(18)I(1)N(6)	426.053

Au+1_OH-1

0	---	3	Au(1)H(1)O(1)	213.977

Au+1_OH-1(2)				
-1	---	5	Au(1)H(2)O(2)	230.985
Au+1_Pen-2				
-1	---	1	C(5)H(9)Au(1)N(1)O(2)S(1)	344.162
Au+1_Pyridine_I-1				
0	---	1	C(5)H(5)Au(1)I(1)N(1)	402.971
Au+1_S2O3-2				
-1	---	1	Au(1)O(3)S(2)	309.088
Au+1_S2O3-2(2)				
-3	---	2	Au(1)O(6)S(4)	421.206
Au+1_SCN-1				
0	---	1	C(1)Au(1)N(1)S(1)	255.048
Au+1_SCN-1(2)				
-1	---	5	C(2)Au(1)N(2)S(2)	313.125
Au+1_SO3-2(2)				
-3	---	2	Au(1)O(6)S(2)	357.086
Au+1_Thiourea				
1	---	3	C(1)H(4)Au(1)N(2)S(1)	273.086
Au+1_Thiourea_I-1				
0	---	2	C(1)H(4)Au(1)I(1)N(2)S(1)	399.986
Au+1_Thiourea_SO3-2				
-1	---	1	C(1)H(4)Au(1)N(2)O(3)S(2)	353.144
Au+1_Thiourea(2)				
1	---	4	C(2)H(8)Au(1)N(4)S(2)	349.202

Au+1_Thiourea(3)

1	---	1	C(3)H(12)Au(1)N(6)S(3)	425.319

Au+1(2)_DMPS-3(2)

-4	---	1	C(6)H(10)Au(2)O(6)S(6)	764.442

Au+1(2)_SCN-1(2)

0	---	1	C(2)Au(2)N(2)S(2)	510.095

Au+1(2)_SCN-1(2)_(s)

0	---	1	C(2)Au(2)N(2)S(2)	510.095

Au+1(3)_AsO4-3_(s)

Gold(I) arsenate; Trigold arsenate

0	---	1	As(1)Au(3)O(4)	729.830

Au+2

Gold(II) ion

2	15456-07-2	5	Au(1)	196.970

Au+2_Met-1

1	---	1	C(5)H(10)Au(1)N(1)O(2)S(1)	345.170

Au+2_Met-1(2)

0	---	1	C(10)H(20)Au(1)N(2)O(4)S(2)	493.370

Au+2_NH3(4)

2	---	1	Au(1)H(12)N(4)	265.092

Au+3

Gold(III) ion

3	16065-91-1	160	Au(1)	196.970

Au+3_Ag+1_F-1(4)_(s)

0	---	1	Ag(1)Au(1)F(4)	380.834

Au+3_Ala-1

2	---	1	C(3)H(6)Au(1)N(1)O(2)	285.056

Au+3_Ala-1(2)

1	---	1	C(6)H(12)Au(1)N(2)O(4)	373.142

Au+3_Asn-1

2	---	1	C(4)H(7)Au(1)N(2)O(3)	328.081

Au+3_Asn-1(2)

1	---	1	C(8)H(14)Au(1)N(4)O(6)	459.192

Au+3_Asn-1(3)

0	---	1	C(12)H(21)Au(1)N(6)O(9)	590.304

Au+3_Asp-2

1	---	2	C(4)H(5)Au(1)N(1)O(4)	328.058

Au+3_Asp-2(2)

-1	---	3	C(8)H(10)Au(1)N(2)O(8)	459.146

Au+3_Asp-2(3)

-3	---	1	C(12)H(15)Au(1)N(3)O(12)	590.234

Au+3_bAla-1

2	---	1	C(3)H(6)Au(1)N(1)O(2)	285.056

Au+3_bAla-1(2)

1	---	1	C(6)H(12)Au(1)N(2)O(4)	373.142

Au+3_Br-1_Cl-1(2)

0	---	2	Au(1)Br(1)Cl(2)	347.780

Au+3_Br-1_Cl-1(3)

-1	---	5	Au(1)Br(1)Cl(3)	383.233

Au+3_Br-1_OH-1(3)

-1	---	1	Au(1)Br(1)H(3)O(3)	327.896

Au+3_Br-1(2)_Cl-1

0	---	2	Au(1)Br(2)Cl(1)	392.231

Au+3_Br-1(2)_Cl-1(2)

-1	---	6	Au(1)Br(2)Cl(2)	427.684

Au+3_Br-1(2)_OH-1(2)

-1	---	1	Au(1)Br(2)H(2)O(2)	390.793

Au+3_Br-1(3)

0	---	2	Au(1)Br(3)	436.682

Au+3_Br-1(3)_(s)

Gold(III) bromide; Gold tribomide; Gold bromide, red-brown; Gold bromide, monoclinic

0	10294-28-7	2	Au(1)Br(3)	436.682

Au+3_Br-1(3)_Cl-1

-1	---	6	Au(1)Br(3)Cl(1)	472.135

Au+3_Br-1(3)_OH-1

-1	---	1	Au(1)Br(3)H(1)O(1)	453.689

Au+3_Br-1(4)

-1	---	11	Au(1)Br(4)	516.586

Au+3_Cl-1_CN-1(3)

-1	---	3	C(3)Au(1)Cl(1)N(3)	310.476

Au+3_Cl-1_OH-1

1	---	3	Au(1)Cl(1)H(1)O(1)	249.430

Au+3_Cl-1_OH-1(3)

-1	---	4	Au(1)Cl(1)H(3)O(3)	283.445

Au+3_Cl-1(2)

1	---	2	Au(1)Cl(2)	267.876

Au+3_Cl-1(2)_OH-1(2)

-1	---	5	Au(1)Cl(2)H(2)O(2)	301.891

Au+3_Cl-1(3)

0	---	4	Au(1)Cl(3)	303.329

Au+3_Cl-1(3)_(s)

Gold(III) chloride; Gold trichloride; Auric chloride

0	---	2	Au(1)Cl(3)	303.329

Au+3_Cl-1(3)_OH-1

-1	---	6	Au(1)Cl(3)H(1)O(1)	320.336

Au+3_Cl-1(4)

-1	---	16	Au(1)Cl(4)	338.782

Au+3_CN-1(3)

0	---	2	C(3)Au(1)N(3)	275.023

Au+3_CN-1(3)_I-1

-1	---	3	C(3)Au(1)I(1)N(3)	401.923

Au+3_Cys-2

1	---	1	C(3)H(5)Au(1)N(1)O(2)S(1)	316.108

Au+3_Dien_Br-1

2	---	1	C(4)H(13)Au(1)Br(1)N(3)	380.041

Au+3_Dien_Cl-1

2	---	3	C(4)H(13)Au(1)Cl(1)N(3)	335.590

Au+3_Dien_N3-1

2	---	1	C(4)H(13)Au(1)N(6)	342.157

Au+3_Dien_SCN-1

2	---	1	C(5)H(13)Au(1)N(4)S(1)	358.215

Au+3_F-1(3)_(s)

Gold(III) fluoride; Gold trifluoride

0	---	2	Au(1)F(3)	253.965

Au+3_Glu-2

1	---	2	C(5)H(7)Au(1)N(1)O(4)	342.085

Au+3_Glu-2(2)

-1	---	3	C(10)H(14)Au(1)N(2)O(8)	487.200

Au+3_Glu-2(3)

-3	---	1	C(15)H(21)Au(1)N(3)O(12)	632.315

Au+3_Gly-1

2	---	1	C(2)H(4)Au(1)N(1)O(2)	271.029

Au+3_Gly-1(2)

1	---	1	C(4)H(8)Au(1)N(2)O(4)	345.089

Au+3_H+1_Br-1(4)_H2O(5)_(s)

0	---	1	Au(1)Br(4)H(11)O(5)	607.670

Au+3_H+1_Cl-1(4)

0	---	2	Au(1)Cl(4)H(1)	339.790

Au+3_H+1_Cl-1(4)_H2O(3)_(s)

Gold chloride trihydrate; Chloroauric acid; Hydrogen tetrachloaurate(III) hydrate; Tetrachloroauric(III) acid

0	16961-25-4	1	Au(1)Cl(4)H(7)O(3)	393.836

Au+3_H+1_Cl-1(4)_H2O(4)_(s)

0	---	1	Au(1)Cl(4)H(9)O(4)	411.851

Au+3_I-1(3)_(s)

0	---	1	Au(1)I(3)	577.670

Au+3_I-1(4)

-1	---	6	Au(1)I(4)	704.570

Au+3_Leu-1

2	---	1	C(6)H(12)Au(1)N(1)O(2)	327.137

Au+3_Leu-1(2)

1	---	1	C(12)H(24)Au(1)N(2)O(4)	457.304

Au+3_Met-1

2	---	1	C(5)H(10)Au(1)N(1)O(2)S(1)	345.170

Au+3_Met-1(2)

1	---	1	C(10)H(20)Au(1)N(2)O(4)S(2)	493.370

Au+3_Na+1_Cl-1(4)

0	---	2	Au(1)Cl(4)Na(1)	361.772

Au+3_NH3(3)

3	---	2	Au(1)H(9)N(3)	248.062

Au+3_NH3(3)_OH-1

2	---	1	Au(1)H(10)N(3)O(1)	265.069

Au+3_NH3(4)

3	---	5	Au(1)H(12)N(4)	265.092

Au+3_NH3(4)_OH-1

2	---	2	Au(1)H(13)N(4)O(1)	282.099

Au+3_OH-1

2	---	5	Au(1)H(1)O(1)	213.977

Au+3_OH-1(2)

1	---	5	Au(1)H(2)O(2)	230.985

Au+3_OH-1(3)

0	---	5	Au(1)H(3)O(3)	247.992

Au+3_OH-1(3)_(s) Gold hydroxide				
0	1303-52-2	8	Au(1)H(3)O(3)	247.992
Au+3_OH-1(4)				
-1	---	11	Au(1)H(4)O(4)	264.999
Au+3_OH-1(5)				
-2	---	5	Au(1)H(5)O(5)	282.007
Au+3_OH-1(6)				
-3	---	1	Au(1)H(6)O(6)	299.014
Au+3_SCN-1(4)				
-1	---	4	C(4)Au(1)N(4)S(4)	429.281
Au+3_SCN-1(5)				
-2	---	2	C(5)Au(1)N(5)S(5)	487.359
Au+3_SCN-1(6)				
-3	---	2	C(6)Au(1)N(6)S(6)	545.436
Au+3_Ser-1				
2	---	1	C(3)H(6)Au(1)N(1)O(3)	301.056
Au+3_Ser-1(2)				
1	---	1	C(6)H(12)Au(1)N(2)O(6)	405.141
Au+3_Thr-1				
2	---	1	C(4)H(8)Au(1)N(1)O(3)	315.082
Au+3_Thr-1(2)				
1	---	1	C(8)H(16)Au(1)N(2)O(6)	433.195
Au+3_Val-1				
2	---	1	C(5)H(10)Au(1)N(1)O(2)	313.110

Au+3_Val-1(2)

1	---	1	C(10)H(20)Au(1)N(2)O(4)	429.250

Au(CH3)2+1
Dimethylgold ion

1	---	1	C(2)H(6)Au(1)	227.040

Au(CH3)2+1(2)_OH-1(2)

0	---	1	C(4)H(14)Au(2)O(2)	488.094

Au(g)
Gold (gas)

0	---	1	Au(1)	196.970

Au(s)
Gold; Gold, cubic

0	7440-57-5	96	Au(1)	196.970

Au2O3(alpha,s)
Gold oxide, alpha; Digold trioxide, alpha

0	1303-58-8	6	Au(2)O(3)	441.938

Au2O3(am.,s)
Gold(III) oxide, amorphous

0	---	3	Au(2)O(3)	441.938

Au2O3(beta,s)
Gold oxide, beta; Digold trioxide, beta

0	---	1	Au(2)O(3)	441.938

AuO2(s)
Gold oxide

0	---	7	Au(1)O(2)	228.969

AuO3-3

-3	---	12	Au(1)O(3)	244.968

AuSe(alpha,s)
Gold monoselenide, alpha

0	---	1	Au(1)Se(1)	275.933

AuSe(beta,s)
Gold monoselenide, beta

0	---	1	Au(1)Se(1)	275.933

AuTe2(s)
Gold ditelluride

0	---	1	Au(1)Te(2)	452.176

bAla-1
beta-Alanate ion

-1	---	498	C(3)H(6)N(1)O(2)	88.0861

Br-1
Bromide ion

-1	24959-67-9	1213	Br(1)	79.9040

Br2(l)
Bromine, liquid

0	---	285	Br(2)	159.808

C(s)
Carbon; Graphite; Carbon, hexagonal

0	7440-44-0	530	C(1)	12.0110

Cl-1
Chloride ion

-1	16887-00-6	2242	Cl(1)	35.4530

Cl2(g)
Chlorine gas

0	7782-50-5	589	Cl(2)	70.9060

CN-1
Cyanide ion

-1	57-12-5	658	C(1)N(1)	26.0177

Cys-2
Cysteinate ion; L-Cysteinate ion; beta-mercaptoalanate ion; 2-amino-3-mercaptopropanoate ion; 2-amino-3-mercaptopropionate ion; alpha-amino-beta-thiolpropionate ion

-2	104170-20-9	673	C(3)H(5)N(1)O(2)S(1)	119.138

DMPS-3
2,3-dimercaptopropane-1-sulphonate ion; unithiolate ion

-3	---	48	C(3)H(5)O(3)S(3)	185.251

e-1
Electron

-1	---	3399	E(1)	0.000000

F-1
Fluoride ion

-1	16984-48-8	1090	F(1)	18.9984

F2(g)
Fluorine gas

0	7782-41-4	259	F(2)	37.9968

Glu-2
Glutamate ion; 2-aminopentanedioate ion; L-Glutamate ion; alpha-aminoglutarate ion; 1-aminopropane-1,3-dicarboxylate ion

-2	138-18-1	650	C(5)H(7)N(1)O(4)	145.115

Gly-1
Glycinate ion; Aminoacetate ion

-1	---	1356	C(2)H(4)N(1)O(2)	74.0593

H+1
Hydrogen ion; Proton

1	12408-02-5	31679	H(1)	1.00794

H+1_AuO3-3

-2	---	5	Au(1)H(1)O(3)	245.976

H+1_CN-1
Hydrogen cyanide, aqueous; Hydrocyanic acid

0	---	23	C(1)H(1)N(1)	27.0256

H+1_Cys-2

-1	---	39	C(3)H(6)N(1)O(2)S(1)	120.146

H+1_NH3
Ammonium ion

1	14798-03-9	108	H(4)N(1)	18.0385

H+1(2)_AuO3-3

-1	---	5	Au(1)H(2)O(3)	246.984

H2(g)
Hydrogen gas; Hydrogen

0	1333-74-0	1564	H(2)	2.01588

H2O
Water

0	7732-18-5	5947	H(2)O(1)	18.0153

I-1
Iodide ion

-1	20461-54-5	894	I(1)	126.900

I2
Iodine, aqueous

0	---	20	I(2)	253.800

I2(s)
Iodine; Iodine, orthorhombic

0	7553-56-2	244	I(2)	253.800

I3-1

-1	---	10	I(3)	380.700

Leu-1
Leucinate ion; L-Leucinate ion; 2-amino-4-methylvalerate ion; alpha-aminoisocaproate ion; 2-amino-4-methylpentanoate ion; L-2-amino-4-methylpentanoate ion

-1	17332-93-3	511	C(6)H(12)N(1)O(2)	130.167

Met-1
Methioninate ion; L-Methioninate ion; 2-amino-4-(methylthio)butyrate ion; alpha-amino-gamma-methylmercaptobutyrate ion; 2-amino-4-methylthiobutanoate ion; L-2-amino-4-(methylthio)butanoate ion; gamma-methylthio-alpha-aminobutyrate ion; Meonine ion; Methilanin ion; Neston ion

-1	93375-49-6	527	C(5)H(10)N(1)O(2)S(1)	148.200

N2(g) Nitrogen gas; Nitrogen				
0	7727-37-9	566	N(2)	28.0134
N3-1 Azide ion				
-1	14343-69-2	176	N(3)	42.0201
Na+1 Sodium(I) ion				
1	17341-25-2	617	Na(1)	22.9900
Na(s) Sodium; Sodium, cubic				
0	7440-23-5	209	Na(1)	22.9900
NH3 Ammonia, aqueous				
0	---	1462	H(3)N(1)	17.0305
O2(g) Oxygen gas; Oxygen				
0	7782-44-7	2559	O(2)	31.9988
OH-1 Hydroxide ion				
-1	14280-30-9	6558	H(1)O(1)	17.0073
Pen-2 Penicillaminate ion				
-2	---	187	C(5)H(9)N(1)O(2)S(1)	147.192
Pyridine Pyridine; Azine; py				
0	110-86-1	297	C(5)H(5)N(1)	79.1014
S(s) Sulphur, alpha; Sulphur, orthorhombic; Sulphur; Sulfur, alpha; Sulfur, orthorhombic; Sulfur; Sulfur-Rhmb				
0	7704-34-9	619	S(1)	32.0600

S2O3-2

Thiosulphate ion; Thiosulfate ion

-2	14383-50-7	276	O(3)S(2)	112.118

SCN-1

Thiocyanate ion; NCS-1 equivalent

-1	302-04-5	785	C(1)N(1)S(1)	58.0777

Se(s)

Selenium, trigonal; Selenium, grey; Selenium, black; Selenium, gamma; Selenium, hexagonal (sic)

0	7782-49-2	140	Se(1)	78.9630

Ser-1

Serinate ion; L-Serinate ion; 2-amino-3-hydroxypropionate ion; beta-hydroxyalanate ion; 2-amino-3-hydroxypropanoate ion; alpha-amino-beta-hydroxypropionate ion; L-2-amino-3-hydroxypropanoate ion

-1	17807-54-4	708	C(3)H(6)N(1)O(3)	104.086

SO3-2

Sulphite ion; Sulfite ion

-2	14265-45-3	146	O(3)S(1)	80.0582

Te(s)

Tellurium; Tellurium, hexagonal

0	13494-80-9	71	Te(1)	127.603

Thiourea

Thiocarbamide; Thiourea; tu

0	62-56-6	261	C(1)H(4)N(2)S(1)	76.1162

Thr-1

Threoninate ion; L-Threoninate ion; 2-amino-3-hydroxybutyrate ion; alpha-amino-beta-hydroxybutyrate ion; 2-amino-3-hydroxybutanoate ion; L-2-amino-3-hydroxybutanoate ion

-1	68199-29-1	602	C(4)H(8)N(1)O(3)	118.112

Val-1

L-Valinate ion; Valinate ion; L-2-Amino-3-methylbutanoate ion

-1	---	582	C(5)H(10)N(1)O(2)	116.140